A uniform Beta integral estimate with a logarithmic-delta corollary

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Editorial note. The counted statement is the author's labelled lemma 'lem:beta estimate' (source lines 1460-1469); the proof environment begins on the next line after \end{lem} apart from one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own lead-in to the section, including the Beta density display and the reflection-formula identity that the lemma's normalisation uses.

Editorial note. No counter adjustment is retained: the publisher numbering is not reproduced and the item prints with the wrapper's own sequential numbering.

Editorial note. No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference.

Editorial scope. One original result of the pinned publisher article, counted once: the uniform Beta integral estimate with its logarithmic-delta corollary (author label lem:beta estimate).

Allowed standard prerequisites (editorial).

- Beta distribution density. If Z follows Dir(alpha) then each component Z_i is Beta(alpha_i, 1 - alpha_i) with density proportional to t^{alpha_i - 1}(1-t)^{-alpha_i}.. It is the display at source lines 1451-1457 that makes the integral of the lemma the quantity to be estimated.
- Euler reflection formula. Gamma(alpha)Gamma(1-alpha) = pi / sin(pi alpha).. It is used at source line 1458 and again inside the proof at line 1473 to normalise the total mass to pi.
- Symmetry of the integrand. The integral and its bound are invariant under (alpha,u) -> (1-alpha, 1-u).. It is the reduction that opens the proof at source line 1471.

Since $\boldsymbol{Z}$ follows the distribution $\mathrm{Dir}(\boldsymbol{\alpha})$, each component Z_i follows a $\mathrm{Beta}(\alpha_i, 1-\alpha_i)$ distribution, meaning that if $[a,b] \subseteq [0,1]$, then

$$\begin{aligned} \mathbb{P}\big[Z_i \in [a,b]\big] &= \frac{1}{\Gamma(\alpha_i)\Gamma(1-\alpha_i)} \int_a^b \frac{\mathrm{d}t}{t^{1-\alpha_i}(1-t)^{\alpha_i}} \\ &= \frac{\sin(\pi\alpha_i)}{\pi} \int_a^b \frac{\mathrm{d}t}{t^{1-\alpha_i}(1-t)^{\alpha_i}}, \end{aligned} \tag{1}$$

by Euler's reflection formula. For this reason, we need the following preliminary estimate.

Lemma 0.1. *Uniformly for $u \in [0,1]$, $\alpha \in (0,1)$ and $\delta > 0$, we have*

$$\sin(\pi\alpha) \int_{[u-\delta,u+\delta]\cap[0,1]} \frac{\mathrm{d}t}{t^{1-\alpha}(1-t)^{\alpha}} \ll \frac{\delta}{(u+\delta)^{1-\alpha}(1-u+\delta)^{\alpha}}.$$

In particular, if $\delta \geq 1/\log x$, then

$$\sin(\pi\alpha) \int_{[u-\delta,u+\delta]\cap[0,1]} \frac{\mathrm{d}t}{t^{1-\alpha}(1-t)^{\alpha}} \ll \frac{\delta \log x}{(1+u\log x)^{1-\alpha}(1+(1-u)\log x)^{\alpha}}.$$

Proof. Note that both sides of the claimed inequality are invariant under the change of variables $(\alpha, u) \mapsto (1-\alpha, 1-u)$. Hence, we may assume without loss of generality that $u \in [0, 1/2]$. In addition, observe that

$$\sin(\pi\alpha) \int_0^1 \frac{\mathrm{d}t}{t^{1-\alpha}(1-t)^{\alpha}} = \frac{\sin(\pi\alpha)}{\Gamma(\alpha)\Gamma(1-\alpha)} = \pi.$$

Hence the lemma is trivially true if $\delta > 1/4$.

We have thus reduced the proof to the case when $\alpha \in (0,1)$, $u \in [0,1/2]$ and $\delta \in [0,1/4]$. In particular, $u+\delta \leq 3/4$, so that $1-t \in [1/4, 1]$ for all $t \in [0, u+\delta]$. It thus suffices to prove that

$$\sin(\pi\alpha) \int_{[u-\delta,u+\delta]\cap[0,1]} \frac{\mathrm{d}t}{t^{1-\alpha}} \ll \frac{\delta}{(u+\delta)^{1-\alpha}}. \tag{2}$$

Assume first that $\delta \leq u/2$. We then have $t \geq u/2$ whenever $t \in [u-\delta, u+\delta]$. Using also the trivial bound $\sin(\pi\alpha) \leq 1$, we conclude that

$$\sin(\pi\alpha) \int_{[u-\delta,u+\delta]\cap[0,1]} \frac{\mathrm{d}t}{t^{1-\alpha}} \leq \int_{[u-\delta,u+\delta]\cap[0,1]} \frac{\mathrm{d}t}{(u/2)^{1-\alpha}} \ll \frac{\delta}{(u+\delta)^{1-\alpha}}.$$

This proves the lemma in this case.

Finally, assume that $\delta \geq u/2$. We then use that

$$\begin{aligned}\sin(\pi\alpha) \int_{[u-\delta,u+\delta]\cap[0,1]} \frac{\mathrm{d}t}{t^{1-\alpha}} &\leq \sin(\pi\alpha) \int_0^{3\delta} \frac{\mathrm{d}t}{t^{1-\alpha}} = \frac{\sin(\pi\alpha)}{\alpha} \cdot (3\delta)^{\alpha} \\ &\ll \delta^{\alpha} \asymp \frac{\delta}{(u+\delta)^{1-\alpha}}.\end{aligned}$$

This completes the proof of the lemma in all cases. □

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